

HACKATHON 2026 • EIRGRID + CSO OPEN DATA

GridSense

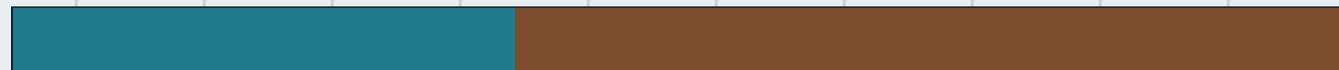
Forecast where Ireland's load will peak.
Route renewables there first. From Kerry to
Belfast.

THE PROBLEM

Ireland's wind is there. It doesn't reach the loads that need it.

37.6%

of island electricity was renewable, Oct 2025 – Oct 2026
(EirGrid, 15-min actuals, ROI + NI)



23.2%

of ROI metered electricity goes to data centres, as flat 24/7
base load (CSO MEC02, 2025)

- On windy nights, wind farms are **dispatched down** because demand isn't there to absorb them.
- At the evening peak, gas fills the gap and the biggest loads run on it.
- Nobody can see **where** renewable power actually goes, by county, by hour.

The grid said no. The rules changed.

- **Jan 2022** EirGrid: no new data-centre connections in Dublin until 2028.
- **Apr 2024** AWS limits capacity in its Dublin region. GPU nodes "maxed out power-wise"; customers pointed to Sweden and other EU regions.
- **Aug 2024** Google's Grange Castle expansion refused over grid capacity (overturned on appeal, Sep 2026).
- **Dec 2025** CRU's new policy: new data centres must bring their own dispatchable generation or storage, and source **80%** of annual demand from new Irish renewables within 6 years.
- **Apr 2026** Amazon backs a €150m Midlands wind farm to power its data centres.

Where GridSense fits

- Operators must now **prove** how renewable their load is. GridSense measures it hour by hour.
- It forecasts where the next peak will hit and books recovered wind for it, easing the peak that triggered the moratorium.
- Councils and CRU get the same per-area numbers.

Sources: RTÉ / DCD, Jan 2022 • The Register via DCD, 10 Apr 2024 • Techerati, Sep 2026 • CRU decision via William Fry, 15 Dec 2025 • Irish Examiner, 27 Apr 2026

Forecast every region's peak. Route renewables to it first.

1 • Forecast regional peaks

Hourly load for each of 45 areas, 7 days ahead, from the trailing 8 weeks of real EirGrid data.

Backtest error **2.8%** ROI • **5.5%** NI



2 • Route renewables to them

Wind that would be dispatched down is scheduled, through network and storage capacity, into the biggest forecast peaks first.

Dublin City **696 MW** Tue 18:00
Cork County → Dublin City **485 MWh**



3 • Prove it, per area

Wind + solar share and CO₂ for any county or district, hour by hour, over 1, 3 or 12 months.

The moat: others report national averages after the fact. GridSense forecasts **where and when** each region will peak, at 2.8% error, and routes renewables there before it happens.

Electrons can't be steered. "Routing" means scheduling transmission, storage and renewable allocation against the forecast, so wind reaches the peak instead of being constrained off.

Same workload. More wind behind it.

- Your load stays exactly as it is. GridSense forecasts your peak and books renewables for it.
- The Dublin–Meath corridor's combined forecast peak: **1,845 MW** this week.
- Over 12 months the corridor gets **299 GWh** of recovered wind at its peak hours, cutting **33.8 kt** CO₂ from a 2.36 Mt footprint.
- The forecast retrains every day, so routing keeps up as your load grows.

What the customer gets

- Hourly-matched evidence for **24/7 carbon-free energy** claims
- Location-based **Scope 2** figures per site
- An answer to CRU's 80% renewable rule, with no change to SLAs

Evidence for where the next wind farm and cable should go.

- Per-county demand growth set against local renewable supply.
- The **Dublin–Meath data-centre corridor** consumes. The **western wind belt** (Kerry, Mayo, Donegal) generates.
- Hard numbers to lobby for more wind and solar, and for grid reinforcement where demand is rising.

Questions it answers

- Which counties' demand is outpacing renewables?
- How much wind is lost to dispatch-down, and where?
- Where will next week's peaks hit, and how much wind can reach them?

Pick the site with the surplus, not the congestion.

- See which regions are net renewable generators: the west and north-west.
- Compare wind + solar share and dispatch-down across candidate counties before committing to a site.
- A site near surplus wind starts out greener and puts less strain on the grid.

For investors

Renewable %, forecast peak load and routing headroom for every county and NI district, in one view.

"How much of my power was renewable this month?"

- Pick your area on the map and see its wind + solar share for the last month, quarter or year.
- See your area's CO₂ footprint as a plain number.
- Next: per-Eircode views from smart-meter data, with consent.

Why it matters

Seeing the number builds trust in the transition and makes flexible tariffs (EV charging, heat pumps) an easy choice.

Observability first. Then measurable cuts.

+621 GWh

dispatched-down wind routed to forecast peaks

71.2 kt

CO₂ avoided

37.6% →

39.1%

island wind + solar share

- Every area gets an hourly CO₂ footprint from EirGrid's real carbon intensity.
- Corporates report against measured hourly data instead of annual averages.

Modelled figures. Real EirGrid demand, wind, solar, CO₂ and wind forecast data (Oct 2025 – Oct 2026). Assumptions: the area split (CS0 MEC03 consumption, ROI; population, NI); dispatch-down = wind forecast minus actual when renewables ≥ 60% (9.2% of ROI wind, close to EirGrid's ~10%); routing recovers 50% of it (the network-constraint share) into the 6 forecast peak hours, weighted by forecast peak².

Open data in, one static dashboard out.

EirGrid API

15-min demand, wind, solar, CO₂ intensity, wind forecast. ROI + NI.

CSO + Census

MECO2 data-centre consumption, MECO3 county consumption, NISRA populations, OSi/OSNI boundaries.



Python model

Area load split, per-area peak forecast, routing of recovered wind from wind areas (by capacity) to the biggest peaks, CO₂ accounting.



Dashboard

Leaflet map of all 45 areas plus Chart.js. A static site with no server.

All public. All real.

SOURCE	WHAT WE USE
EirGrid Smart Grid Dashboard API	15-min demand, wind, solar, CO ₂ intensity, wind forecast. ROI + NI, 12 months
CSO PxStat MEC02	Data-centre share of metered electricity, quarterly, 2015–2025
CSO PxStat MEC03	Metered consumption by county and Dublin postal district, 2025
NISRA	Population by NI local government district
OSi / OSNI	County and LGD boundaries

Next data

- ESB Networks smart-meter data per Eircode, with consent
- EirGrid connected wind-farm list per node
- SEAI energy statistics

PROTOTYPE

Live demo

- Read the **Highest forecast load** ranking: where next week's peaks hit
- Click **Dublin City**, then **Fingal**, Ireland's data-centre hub
- Toggle **With GridSense routing** and switch to **12 months**
- Hover a routing line: **Cork County → Dublin City, 485 MWh** this week
- Read **Where the routed wind comes from**: Cork County 1.3 GWh, Kerry 1.3, Donegal 1.0; 12.8 GWh planned

From open data to live routing.

- **Now:** island-wide observability on real EirGrid data
- **Next:** per-Eircode load forecasts from ESB Networks smart meters
- **Then:** live day-ahead routing plans with EirGrid and SONI
- **Later:** hourly certificates for 24/7 CFE and Scope 2 reporting
- **Add-on:** flex signals for customers who want to go further

The ask

- Node-level wind data from EirGrid and SONI
- A data-centre pilot partner in Fingal or Meath
- A policy sponsor for forecast-led grid scheduling

Two bets beyond the prototype.

1 • Green-hour tokens

Rate-limit the island's forecast renewable output into hourly tokens, the same way an API rations requests.

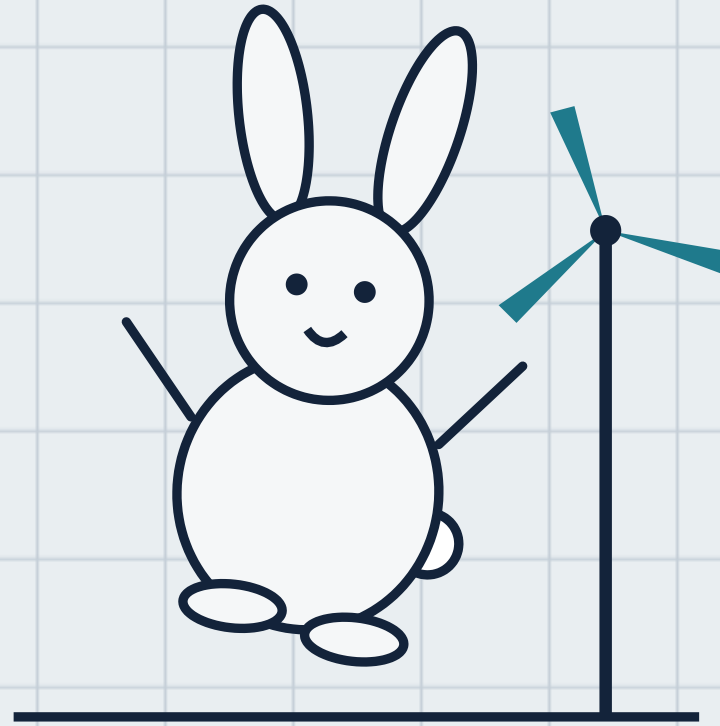
- Companies earn **priority access** to green hours by funding new renewables.
- A per-company cap stops any one buyer from cornering the wind.
- Unused tokens go back to the pool for everyone else.
- Every token spent is an auditable hourly record for CSR and Scope 2 reports.

2 • Micro-turbines on Ireland's canals

The Grand, Royal, Barrow and Shannon–Erne waterways and their feeders were built to manage water flow. That flow can turn small turbines too.

- Fit small turbines at locks, weirs and sluices, where the water drops.
- Unlike wind, the output is steady day and night.
- GridSense picks sites where local generation is lowest compared with demand.
- Partners: Waterways Ireland, local councils and community energy groups.

Canal water moves slowly, so most of the energy is at the drops (locks and weirs), not in the open channel. A pilot at one lock would show the real yield.



Thank you.

This bunny's moves are powered by wind
that would otherwise have been switched off.

Questions?